

IOE 611/ Math 663

Lecture 1: Introduction

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IOE 611/Math 663: Nonlinear Programming

- ▶ **Lectures:** Tue and Thu, 12:00–1:30 PM, 1005 Dow
- ▶ **Instructors:**
 - ▶ Salar Fattahi
OH: Tuesdays 4:00-5:00PM, Wednesdays 9:00-10:00AM
 - ▶ Geyu Liang
OH: Mondays 1:30:00-2:30PM, Fridays 12:30-1:30PM
 - ▶ OH will be on Zoom.
- ▶ **Main Textbook:** “Convex Optimization” by Stephen Boyd and Lieven Vandenbergh
<https://web.stanford.edu/~boyd/cvxbook/>
See syllabus for a complete list of references.
- ▶ **Required background:** Working knowledge of linear algebra and analysis (review Appendix A of the book). Also, a level of mathematical maturity to program in Matlab. Exposure to numerical computing, optimization, and its application fields is helpful.

Please read syllabus for the complete course policies.

Course Logistics

- ▶ Lectures:
 - ▶ Slides posted prior to lectures
 - ▶ In class, most mathematical proofs will be handwritten.
 - ▶ The annotated slides will be also posted.
- ▶ Problem sets every two weeks or so, include some computer modeling and programming (35% total)
- ▶ Two take-home exams (30% each)
- ▶ Active participation (5% total)
- ▶ Written solutions must be typed, $\text{\LaTeX}$ highly recommended

Covid policies

- ▶ If you feel sick, do not come in; the lectures will be streamed live, and will be recorded.
- ▶ No eating or drinking in class!

Mathematical optimization

(mathematical) optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq b_i, \quad i = 1, \dots, m\end{array}$$

- ▶ $x = (x_1, \dots, x_n)$: optimization variables (or decision variables)
- ▶ $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}$: objective function
- ▶ $f_i : \mathbf{R}^n \rightarrow \mathbf{R}, i = 1, \dots, m$: constraint functions

optimal solution x^* has smallest value of f_0 among all vectors that are *feasible*, i.e., satisfy the constraints

The high level overview of this course is about solving an optimization problem in the above form. we want to minimize a function $f_0(x)$, x can be scalar, vector, can be matrix, or even can be tensor.

$f_0(x)$ - function we are trying to minimize we call it objective function.

$$f_0: \mathbb{R}^n \rightarrow \mathbb{R} \quad (\text{we only work with Real numbers})$$

$\Rightarrow$ By minimization means we want to find an x that leads to the smallest possible value for $f_0(x)$.

x = decision variable, optimization variable
 x can be scalar, vector, matrix or tensor

*in this course
we won't work*

But we also have constraints

$f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, f_i defines set of

constraint's that we need to satisfy. $f_i(x) \leq b_i$ means any solⁿ that we obtain by solving this optimization problem should satisfy this inequality and we can have as many inequality constraint's we want. These kind of constraint's can come from Law's of Physics, Budget, anything can be encoded as constraint's.

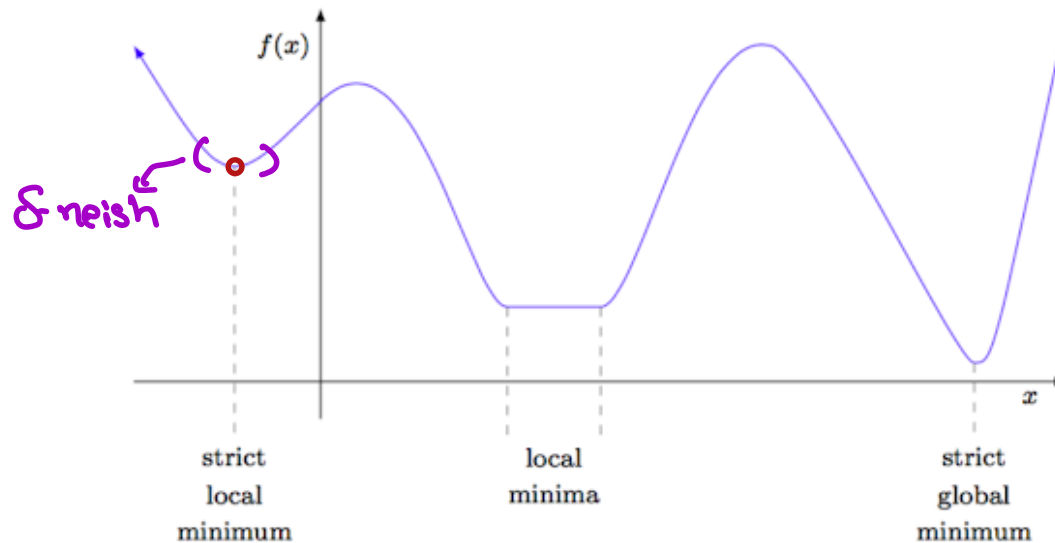
* our main goal is to find optimal solⁿ. x^* is the optimal solⁿ if it has smallest value of $f_0(x)$ among all the point's (vector's, matrix's etc) that are feasible (satisfy the constraint's)

in this problem f_0 , f_i , b_i are given, we just need to find x^* , The only thing we don't know is x^* . and we are trying to find x .

* There are different tricks that need to be used to write it as a well defined optimization problem, and what does it mean by well defined? Really when you write an optimization problem at the end of the day what we are trying to do is to solve, we want to give it to a solver and get an optimal solⁿ or maybe near optimal solⁿ. So if we want to give it to solver it should be well defined.

Local/Global solutions

- ▶ A point x is a **local minimum** of $f(x)$ if there exists $\delta > 0$ such that $f(x) \leq f(y)$ for every y such that $\|x - y\| \leq \delta$.
- ▶ A point x is a **global minimum** of $f(x)$ if $f(x) \leq f(y)$ for every y .



A point x is a local minimum of $f(x)$, if it's best possible solⁿ within a neighborhood of x

x is local if $\exists \delta > 0$ s.t. $f(x) \leq f(y)$
 $\forall y$ (for every y) within Ball of
 x ($\underbrace{\|x - y\|}_{\text{Norm (can be } L_2, L_1, F\text{-Norm etc)}}$ $\leq \delta$)

δ is also arbitrary $\Rightarrow$ can be as small as possible as long as it's positive.

Strict local minimum $\Rightarrow f(x) < f(y)$ $\forall y$ in the δ -neighborhood of $x \Rightarrow \|x - y\| \leq \delta$ ($y \neq x$)

Global min: There is no other point that would give us better objective value. (smaller objective value)

Simillary Strict Global min: $f(x) < f(y)$
 $\forall y \in \text{feasible set}$
& $y \neq x$

Examples (one of the most commonly used Problem)

portfolio optimization: looking for a best way to invest in a set of n assets

- ▶ variables: amounts invested in different assets
- ▶ constraints: budget, max./min. investment per asset, minimum return
- ▶ objective: risk

n - different options, & certain amount of money
we want to invest in those assets.

Examples

portfolio optimization: looking for a best way to invest in a set of n assets

- ▶ variables: amounts invested in different assets
- ▶ constraints: budget, max./min. investment per asset, minimum return
- ▶ objective: risk

data fitting: from a family of potential models, find a model that best fits observed data and prior information

- ▶ variables: model parameters
- ▶ constraints: prior information, parameter limits
- ▶ objective: measure of misfit or estimation error

Solving optimization problems

solution method

- ▶ A *solution method* for a class of problems is an algorithm that computes the solution to a given *accuracy*, given an instance of a problem from the class.

most difficult part of solving any optimization problem is to come up with a solution method or an algorithm.

what is a solution method or algorithm, is that compute's solution to a given accuracy $\rightarrow$ given an instance of a problem from the class

There are two things that we need to keep in mind

① Class of Problem's

② Given accuracy

in many cases we cannot find ^{the} globally optimal solⁿ, But we can find the globally optimal solⁿ upto certain accuracy. for example to machine

precision. and the other thing in this class of Problem's should be ideally as large as possible

Ex: Class of Problem's that we want to work with is quadratic function's $ax^2 + bx + c$

Ex: $\min x^2$

Solution method: return 0 } But for $\min ax^2$
0 is not optimal,

really what we are looking for is an algorithm that gives us the global optimal solⁿ for a class of optimization Problem's not for specific Problem. That's the main difficulty in solving an optimization Problem.

Solving optimization problems

solution method

- ▶ A *solution method* for a class of problems is an algorithm that computes the solution to a given *accuracy*, given an instance of a problem from the class.

general optimization problem

- ▶ very difficult to solve
- ▶ methods involve some compromise, e.g., very long computation time, or not always finding the solution

we are gonna look at different classes of optimization problems that we can solve or obtain solution method efficiently.

in general finding a efficient solⁿ method is not necessarily easy, in fact it's very difficult. we can imagine if the class of problems we are trying to solve is very large obviously this can be very difficult problem, what do we mean by difficult? sometimes we have to make compromises, if we want to find the globally optimal solⁿ, we may need run our algorithm for longer time (days, years)

* But if we don't care much about the quality of our solⁿ, we can reduce runtime of algorithm, that's kind of compromise we typically make in optimization problems.

Least-squares (1st class of optimization problems)

$$x \in \mathbb{R}^n$$

$$\text{minimize } \|Ax - b\|_2^2$$

$$\leadsto \|x\|_2^2 = x^T x$$

$$A \in \mathbf{R}^{k \times n}, \quad k \geq n, \quad \text{rank}(A) = n$$

$$\begin{aligned} \Rightarrow \|Ax - b\|_2^2 &= (Ax - b)^T (Ax - b) \\ &= (x^T A^T - b^T) (Ax - b) \\ &= \underbrace{x^T A^T A x - b^T A x - x^T A^T b + b^T b}_{\text{quadratic function of } x} \end{aligned}$$

from steel analysis to find min, what to do?

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = 0 \Rightarrow 2A^T A x - 2A^T b = 0$$
$$\Rightarrow x^* = (A^T A)^{-1} A^T b$$

Least-squares

$$x^* = (A^T A)^{-1} A^T b \quad \text{we have closed}$$

form solⁿ. we know $A, b \Rightarrow$ find x^* .

$$\text{minimize } \|Ax - b\|_2^2$$

$$A \in \mathbf{R}^{k \times n}, \quad k \geq n, \quad \text{rank}(A) = n$$

solving least-squares problems

► analytical solution: $x^* = (A^T A)^{-1} A^T b$

$$ATA \approx n^2 k \text{ flops}$$

Least-squares

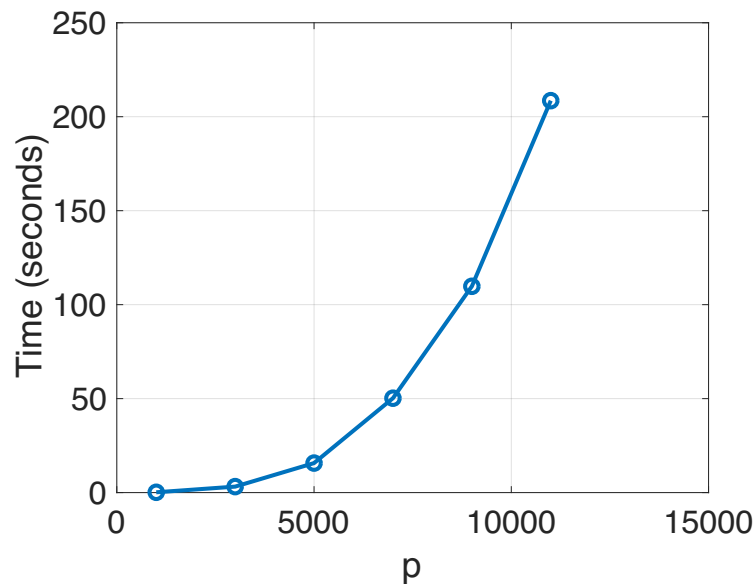
$$\text{minimize } \|Ax - b\|_2^2$$

$$A \in \mathbf{R}^{k \times n}, \quad k \geq n, \quad \text{rank}(A) = n$$

solving least-squares problems

► analytical solution: $x^* = (A^T A)^{-1} A^T b$

$(A^T A)^{-1}$ exists if $\text{Rank}(A) = n$
column rank



Linear programming (another class, we know how to solve)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

two main assumption's

- ① Objective should be linear $\Rightarrow f_0(x)$
- ② Constraints should be linear $\Rightarrow f_i(x)$

Linear programming

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

solving linear programs

- ▶ no analytical formula for solution
- ▶ reliable and efficient algorithms and software
- ▶ computation time *in practice* proportional to $n^2 m$ if $m \geq n$; less with structure
- ▶ a “mature technology”

$\Rightarrow$ we have no closed form sol'n, for linear programs, we have no choice but to use iterative methods to solve linear programs. (But it's a very mature class of optimization problems) we know exactly how to solve them, algorithms we have very reliable

Linear programming

developed 1940-1950: WWII

(Simplex: well known algorithm for linear programming)

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

solving linear programs

(Dantzig Simplex: 1948)

- ▶ no analytical formula for solution
- ▶ reliable and efficient algorithms and software
- ▶ computation time *in practice* proportional to $n^2 m$ if $m \geq n$; less with structure
- ▶ a “mature technology”

(Karmarkar developed Interior Point 1984)

using linear programming

- ▶ not as easy to recognize as least-squares problems
- ▶ a few standard tricks used to convert problems into linear programs (e.g., problems involving ℓ_1 - or ℓ_∞ -norms, convex piecewise-linear functions)

Convex optimization problem (Largest, & Perhaps the most modern class of optimization problems that are considered easy)

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq b_i, \quad i = 1, \dots, m\end{array}$$

- objective and constraint functions are **convex**: for $i = 0, 1, \dots, m$,

$$f_i(\alpha x + \beta y) \leq \alpha f_i(x) + \beta f_i(y)$$

for any $x, y \in \mathbf{R}^n$ and any $\alpha \geq 0, \beta \geq 0$ such that $\alpha + \beta = 1$

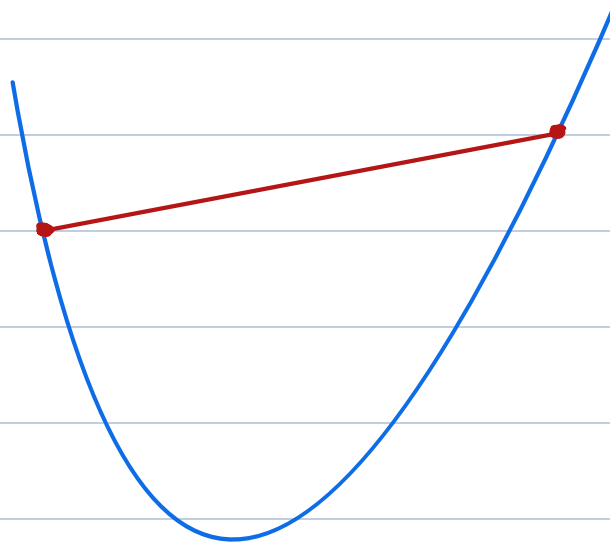
- includes least-squares problems and linear programs as special cases

The Largest, Perhaps the most modern class of optimization problems that are considered easy in the set of convex optimization problems. (in fact main focus of the course)

- * we will try to develop a theory behind convex optimization problems, and algorithms for solving convex optimization problems.

informal & intuitive definition Convex optimization

Convex function: $f(x)$ is convex if we pick any two points on the function, the line segment between these two points lies above the function. Any bowl shape function is a convex.



This class of optimization problems, when both objective function, & constraint's are convex function (f_0, f_i convex), this is considered as convex optimization problem.

Turn's out that it is easy to solve, meaning that there are class of algorithm's that can solve optimization problems like these fairly efficiently.

Convex optimization problem

solving convex optimization problems

- ▶ no analytical solution
- ▶ reliable and (sometimes) efficient algorithms

“In fact the great watershed in optimization isn’t between linearity and nonlinearity, but convexity and nonconvexity.”

—Rockafellar 1993

When we started develop theory for convex optimization
lot of researcher's Back in 1990 realize that the
real question is not about linearity & non-
linearity. The real question is whether
we have convex objective & constraints
or not.

Convex optimization problem

solving convex optimization problems

- ▶ no analytical solution
- ▶ reliable and (sometimes) efficient algorithms

“In fact the great watershed in optimization isn’t between linearity and nonlinearity, but convexity and nonconvexity.”

—Rockafellar 1993

using convex optimization

- ▶ often difficult to recognize
- ▶ many tricks for transforming problems into convex form
- ▶ Convex relaxation techniques

Convex Optimization Problem's are not easy to recognize, in fact we need to use series of tricks to be able to convert an Optimization Problem into Convex Optimization Problem. (learn later in course)

* even if the Problem non-convex in many cases we can actually relax the Problem into Convex Optimization Problem.

* we will see one or 2 these kind of Problem's where the Problem is not necessarily convex, But there are ways to convert them or to actually to at least relax them into a Convex Optimization Problem.

Nonconvex (nonlinear) optimization

traditional techniques for general nonconvex problems involve compromises

The Problem suddenly becomes very difficult to solve.
we use generally local search algorithm.

Nonconvex (nonlinear) optimization

traditional techniques for general nonconvex problems involve compromises

local methods

- ▶ find a point that minimizes f_0 among feasible points near it, useful if goal is to find a good point
- ▶ fast, can handle large problems
- ▶ require initial guess
- ▶ provide no information about distance to (global) optimum

Nonconvex (nonlinear) optimization

traditional techniques for general nonconvex problems involve compromises

local methods

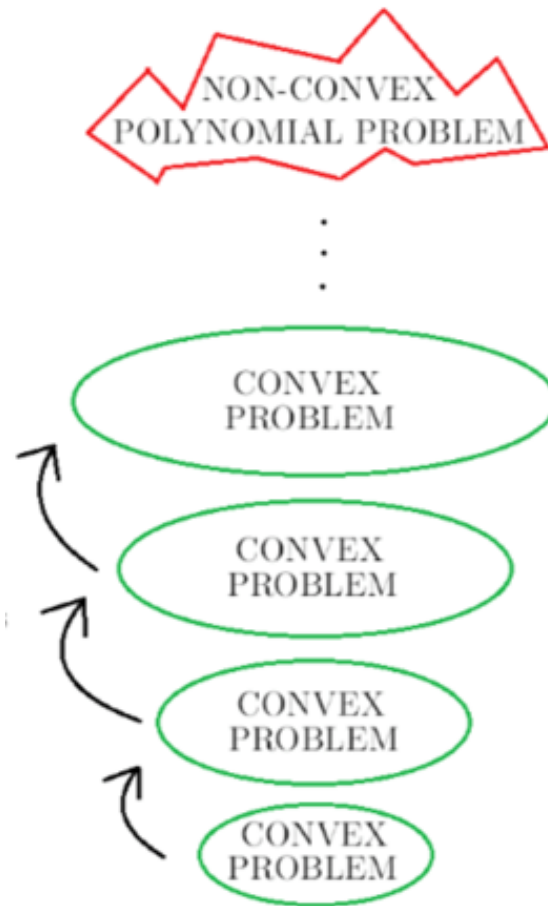
- ▶ find a point that minimizes f_0 among feasible points near it, useful if goal is to find a good point
- ▶ fast, can handle large problems
- ▶ require initial guess
- ▶ provide no information about distance to (global) optimum

global methods

- ▶ find the (global) solution
- ▶ worst-case complexity grows exponentially with problem size

these algorithms are often based on solving convex subproblems

Convex relaxation



Course goals and topics

Major Part of course is to study the theoretical foundation's of convex optimization solution's

goals

1. recognize/formulate problems as convex optimization problems
2. develop code for problems of moderate size
3. characterize optimal solution, give limits of performance, etc.

topics

1. Convex optimization
2. Numerical algorithms
3. Nonconvex optimization

we are not gonna develop solver to solve C.V Problem's.

Convex Sets

The 1st step to understand convex optimization is to understand convex sets. Properties of C.S etc.

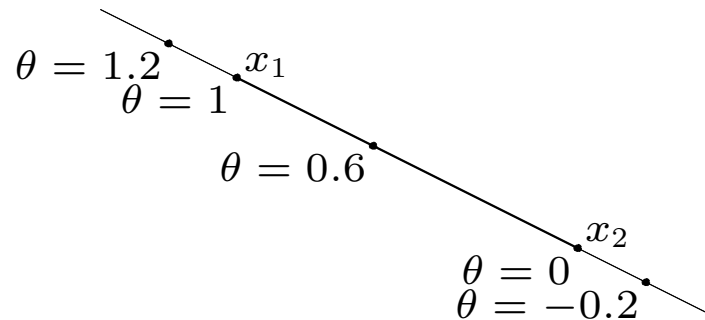
Outline

- ▶ Affine and convex sets, and convex cones
- ▶ Some examples
- ▶ Polyhedron
- ▶ Positive semidefinite cone

Affine set

affine combination of x_1 and x_2 : any point x of the form

$$x = \theta x_1 + (1 - \theta)x_2 \quad \text{where } \theta \in \mathbf{R}$$



line through x_1, x_2 : all affine combinations of x_1 and x_2

x_1, x_2 can be scalar's, vector's, or matrices. But they need to have the same size.

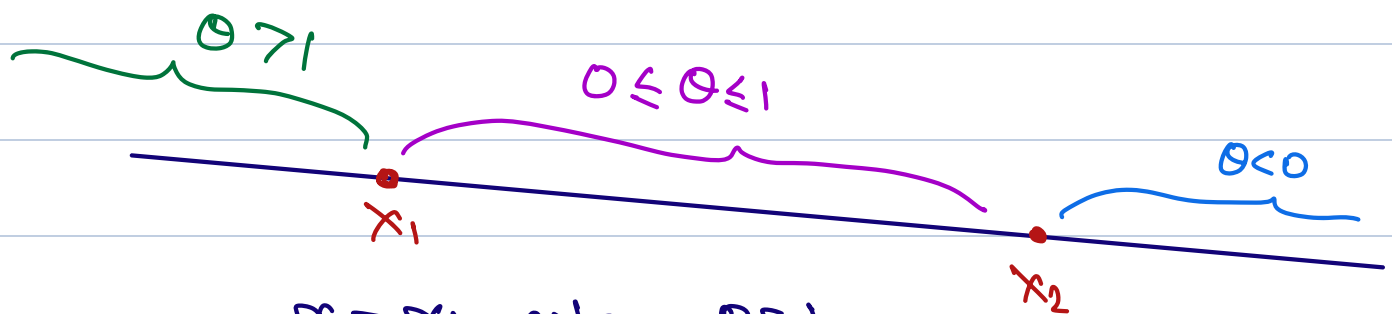
The Affine combination of x_1, x_2 is any point x of the above form.

$$x = \theta x_1 + (1 - \theta)x_1, \quad \theta \text{ should be scalar}$$

$x_1, x_2 \in \mathbb{R}$ or $\mathbb{R}^n$, or $\mathbb{R}^{m \times n}$ anything
But θ should be scalar. $\Rightarrow \theta \in \mathbb{R}$

intuitively the points that characterize like this are
the points that lie on the line that goes through
 x_1 & x_2 .

if θ is b/w 0 & 1 i.e. if $\theta \in (0,1)$, then that
characterization covers the segment b/w x_1
& x_2 .



$x = x_1$ when $\theta = 1$

$x = x_2$ when $\theta = 0$

So really intuitive definition of affine combination
of two points is the line that passes through those
points.

But the definition can be extended to k
points, doesn't have to be 2 points.

if $x_1, x_2, \dots, x_k$ The affine combination of these k points is any x that can be written as

$$x = \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_k x_k$$

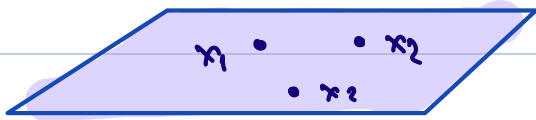
with one condition

$$\alpha_1 + \alpha_2 + \dots + \alpha_k = 1$$

definition of affine combination of k -points

Based on this definition we can define affine sets.

Ex:



Plane passing through x_1, x_2, x_3
 $x \in \mathbb{R}^2$

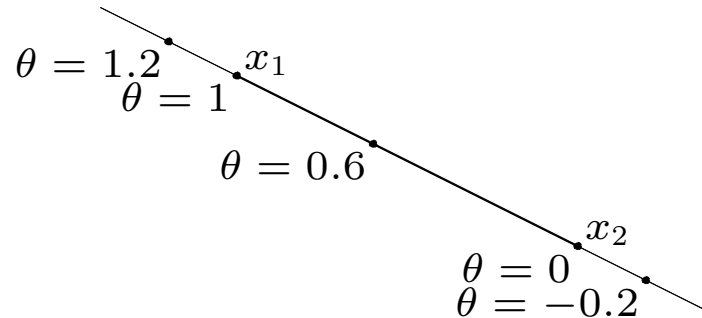
A set is Affine if it contains the line through any two distinct points in the set.

* if I give you a set S and you pick any two points in the set, if the line that goes through those two points is also in the set, that set is called Affine.

Affine set

affine combination of x_1 and x_2 : any point x of the form

$$x = \theta x_1 + (1 - \theta)x_2 \quad \text{where } \theta \in \mathbf{R}$$



line through x_1, x_2 : all affine combinations of x_1 and x_2

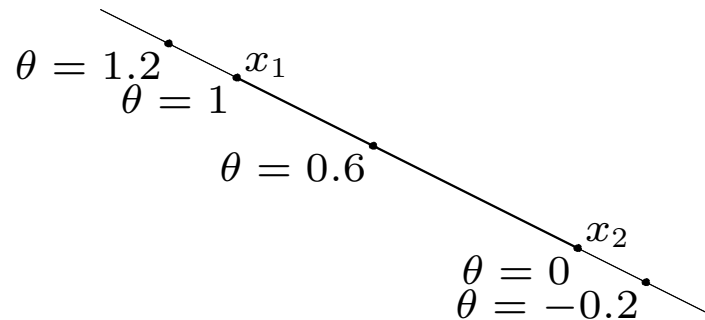
affine set: contains the line through any two distinct points in the set (i.e., a set that's closed under affine combinations)

affine combination

Affine set

affine combination of x_1 and x_2 : any point x of the form

$$x = \theta x_1 + (1 - \theta)x_2 \quad \text{where } \theta \in \mathbf{R}$$



line through x_1, x_2 : all affine combinations of x_1 and x_2

affine set: contains the line through any two distinct points in the set (i.e., a set that's closed under affine combinations)

example: solution set of linear equations $\{x \mid Ax = b\}$

Example: The solution set of $Ax=b$ is an affine set

Solⁿ $C = \{x \mid Ax=b\}$

Pick any two points $x_1, x_2 \in C$, the

line that passing through the points should belong to set C

$$\begin{aligned} x_1, x_2 \in C &\Rightarrow Ax_1 = b && \times 0 \\ & && Ax_2 = b && \times (1-0) \end{aligned}$$

$$\begin{aligned} \Rightarrow 0x_1 &= 0b \\ A(1-0)x_2 &= (1-0)b \end{aligned}$$

$$\Rightarrow A(0x_1 + (1-0)x_2) = 0b + (1-0)b$$

$$\Rightarrow A(\underbrace{0x_1 + (1-0)x_2}_{\in C}) = b$$

Hence $\{x \mid Ax=b\}$ is Affine.

if we can Prove Affine combination for two points, then we can prove it for more than two points, for any k .
(by Induction)

geometric expression?

if we have more than two points, the flexibility

is very limited. Any Affine set can be written in the above form $(Ax=b)$. Any Affine set (solution set) for a bunch of linear equalities.

in higher dimension, Affine sets are subspaces.
(hyper plane, line)

Affine set can always be written as solution set of linear inequalities $\{x \mid Ax=b\}$

Convex sets

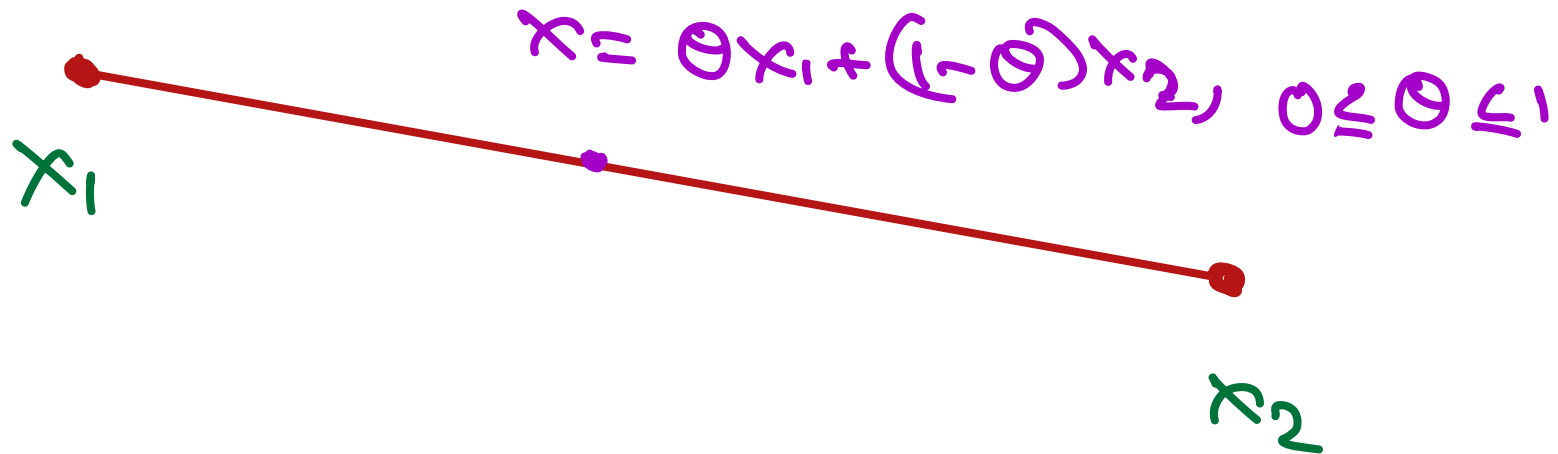
convex combination of x_1 and x_2 : any point x of the form

$$x = \theta x_1 + (1 - \theta)x_2 \text{ where } 0 \leq \theta \leq 1$$

line segment between x_1 and x_2 : all points

$$x = \theta x_1 + (1 - \theta)x_2$$

with $0 \leq \theta \leq 1$



Convex Combination of two points x_1 & x_2 ,
the definition is very similar to affine
combination with one small difference.
The difference is $0 \leq \theta \leq 1$ (θ is restricted)

for Affine Combination $\Rightarrow \theta \in \mathbb{R}$

Convex Combination $\Rightarrow \theta \in [0, 1]$

intuitively the convex combination of two points
is the segment b/w x_1 & x_2 . It's not
a line, it's a segment.

Convex sets

convex combination of x_1 and x_2 : any point x of the form

$$x = \theta x_1 + (1 - \theta)x_2 \text{ where } 0 \leq \theta \leq 1$$

line segment between x_1 and x_2 : all points

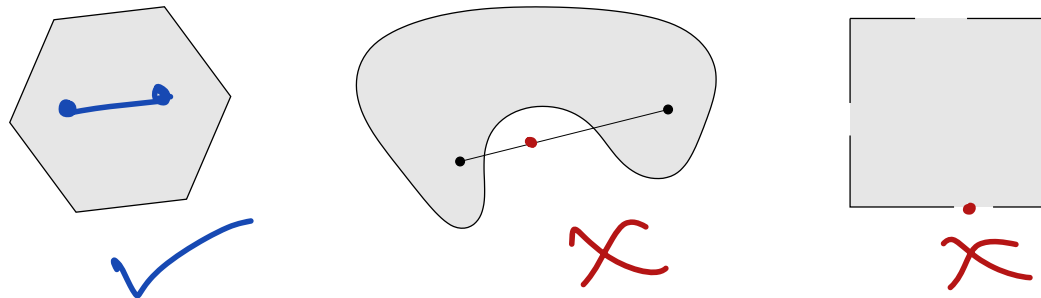
$$x = \theta x_1 + (1 - \theta)x_2$$

with $0 \leq \theta \leq 1$

convex set: contains line segment between any two points in the set (i.e., closed under convex combinations)

$$x_1, x_2 \in C, \quad 0 \leq \theta \leq 1 \quad \implies \quad \theta x_1 + (1 - \theta)x_2 \in C$$

examples (one convex, two nonconvex sets)



Convex set: A set is convex set if we pick any two points on that set that segment b/w those two points should be also in the set. or Convex combination of those two points should also belong to the set.

$$x_1, x_2 \in C, 0 \leq \theta \leq 1 \Rightarrow \theta x_1 + (1-\theta)x_2 \in C$$

This is one of the ways to prove convexity of the set directly using the definition.

Affine sets are convex?

Affine set $\implies$ convex

$\nleftarrow$
not always

Again convex combination can be extended to more than two points exactly same way as Affine combination.

Convex combination and convex hull

convex combination of $x_1, \dots, x_k$: any point x of the form

$$x = \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k$$

with $\theta_1 + \dots + \theta_k = 1, \theta_i \geq 0$ (two conditions)

Corollary:

A set is convex iff $\iff$ it contains any convex combination of any finite number of its points.

Proof: one direction ($\Leftarrow$)

if the set contains any convex combination of any finite number of its points
the claim is the set is convex.

any finite number $\Rightarrow$ I will take 2
(definition)

Hence done.

($\Rightarrow$) Homework

(Hint: use induction) Assume it holds
for k points $\geq$ Prove it for $k+1$ points

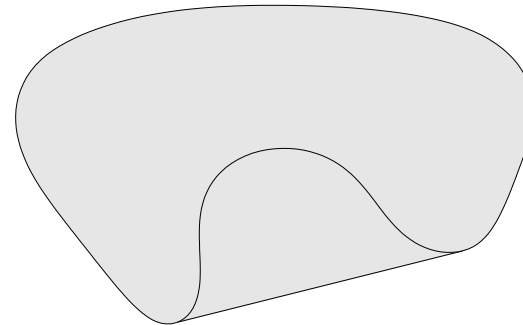
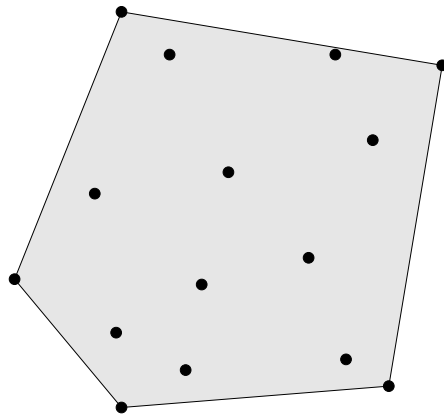
Convex combination and convex hull

convex combination of $x_1, \dots, x_k$: any point x of the form

$$x = \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k$$

with $\theta_1 + \dots + \theta_k = 1$, $\theta_i \geq 0$

convex hull $\text{conv } S$: set of all convex combinations of points in S



Suppose I give you set S [Bunch of points]

that set S does not have to be convex.
can be discrete set,

The set of all convex combination of the points in S is called convex hull of S , we write it as $\text{conv } S$

($\because S$ does not have to be convex set)

①) is convex hull a convex set? **yes**

① Convex hull by definition is a convex set.

② Convex hull of a set S is the smallest convex set that contains S .

② Convex hull of a set S is the smallest convex set that contains S

$D = \text{Convex Set}$

By contradiction suppose $\exists D : S \subseteq D$
 $\text{conv} S \not\subseteq D$

$\Rightarrow \exists x \in \text{conv} S$ s.t. $x \notin D$

$$x = \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k$$

where $x_1, x_2, \dots, x_k \in S$

$$\theta_1 + \theta_2 + \dots + \theta_k = 1$$

$x_1, x_2, \dots, x_k \in S$

$\Rightarrow x_1, x_2, \dots, x_k \in D$ (because $S \subseteq D$)

$\&$ D is a convex set

$\Rightarrow \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k \in D$

(contradiction)